AUDIT REPORT

order-outcome-margin: demo video audit report

A slide-by-slide record of every number, assumption and calculation in the demo video, with its source.

Item	Value
Repository	https://github.com/kavya2693/order-outcome-margin
Numbers taken from commit	`e78d53a391889af780be9f07f440398e04edc426` (outputs last changed in `e78d53a (2026-10-13 22:05:19 +0400)`)
Video	`docs/demo/demo.mp4` · 85.6 s · 1920x1080 · 30 fps · silent
Generated	2026-10-14 14:43
Automated checks	57 pass · 0 fail · 3 manual (of 60 figures)
Assumptions	11 (A-01 to A-11)

How to use this report

Every number or claim on screen has an ID (F-01, F-02, ...) and every modelling assumption has an ID (A-01, ...). Cite the ID. Each figure lists its source file, the exact column, row or key, the formula for derived numbers, and an automated check result. The change-impact matrix (section 5) shows which slides and files change if an input or assumption changes.

Types: direct = read straight from a repo output; derived = computed from direct values (formula given); illustrative = drawing or example, not data; general knowledge = context not produced by the repo.

Regenerate

```
make run                                    # data/, outputs/*.csv, results.json, report.md, *.png (about 2 min)
python3 docs/demo/source/tools/darkmap.py outputs docs/demo/source/assets
PYTHONPATH=src .venv/bin/python docs/demo/source/tools/export_cod_sheet.py docs/demo/source/assets/cod_sheet.json
python3 docs/demo/source/tools/inject_sheet.py docs/demo/source/assets/cod_sheet.json docs/demo/source/compositions/value.html --rows 315400,316055,316096,316229,315744,316309 --animate 315400,316096,316229,315744
cd docs/demo/source && npx --yes hyperframes@0.8.156 check && npx --yes hyperframes@0.8.156 render -f 30 -q looks -o renders/order-outcome-margin-master.mp4 && cd -
ffmpeg -y -i docs/demo/source/renders/order-outcome-margin-master.mp4 -c:v libx264 -preset slow -crf 25 -pix_fmt yuv420p -movflags +faststart -an docs/demo/demo.mp4
python3 docs/demo/source/tools/build_audit_report.py docs/demo/source/audit/facts.yaml   # stills, checks, MD/DOCX/PDF
```

1. Figures

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
F-01	3 chips	01	author's framing; README 'The problem' (finance and operations teams)	-	-	general knowledge	manual: Context statement, not a repo output.
F-02	4 problem cards	01	README 'The problem'	README lines 12-25	-	general knowledge	✓ verified
F-03	341,880 orders	02, 05	outputs/results.json	orders_total	-	direct	✓ verified
F-04	2 years / 2024–2025	02, 05	src/order_margin/config.py:70-71	DATA_START, DATA_END	-	direct	✓ verified
F-05	48%	02	data/orders.csv.gz	payment == 'cod'	share of COD orders = 48.3%	derived	✓ verified
F-06	Five ways each one can end / 5 ways it could end	02, 03	src/order_margin/config.py:11	CLASSES	len = 5	direct	✓ verified
F-07	~470 orders a day	02	data/orders.csv.gz	orders / distinct order days	341,880 / 731 = 467.7, rounded to tens (README: 'around 470 orders a day')	derived	✓ verified
F-08	6 delivery zones, 2 remote	02	src/order_margin/config.py:17-18	ZONES, REMOTE_ZONES	-	direct	✓ verified
F-09	1 in 5 orders is not delivered	02	data/orders.csv.gz	outcome != delivered	1 - 0.798 = 20.2% = 1 in 5 (test window: 20.9%)	derived	✓ verified
F-10	scored as cash and as card	03	src/order_margin/pipeline.py:193-201	with_payment(Xt_cod, 'card')	-	direct	✓ verified
F-11	daily margin / cash rules / beats risk rule	03	outputs/daily_margin.csv, outputs/results.json	122 daily rows; margin_minus_risk_rule > 0	-	derived	✓ verified
F-12	CALIBRATE · make the odds honest	03	src/order_margin/calibration.py:26-39	IsotonicCalibrator	-	direct	✓ verified
F-13	+22,258	04, 06, 08	outputs/policy.csv	policy='margin rule, gbm + isotonic', margin_change	-	direct	✓ verified
F-14	−2,113	04, 06, 08	outputs/policy.csv	policy='risk score, same block rate as margin rule on calib (14.6%)', margin_change	-	direct	✓ verified
F-15	config.py, generate.py, features.py, models.py, calibration.py, economics.py, policy.py, pipeline.py, data/truth	05	src/order_margin/	classes and functions	-	direct	✓ verified
F-16	23 columns	05	src/order_margin/features.py:21-45	NUMERIC (20) + CATEGORICAL (3)	20 + 3 = 23	direct	✓ verified
F-17	history ≥14 d old	05	src/order_margin/features.py:15	RESOLVE_DAYS = 14	-	direct	✓ verified
F-18	186 rounds	05	outputs/results.json	gbm_rounds	-	direct	✓ verified
F-19	× 5 trees / 5-outcome softmax	05	src/order_margin/models.py:74-75	objective multiclass, num_class = N_CLASSES = 5	186 rounds x 5 = 930 trees	derived	✓ verified
F-20	fit May–Aug 25	05	src/order_margin/config.py:68-69	EARLY_STOP_END 2025-04-30, CALIB_END 2025-08-31	-	direct	✓ verified
F-21	40% / 60% · assumed input	05	src/order_margin/config.py:75	COD_TO_CARD_CONVERSION	-	direct	✓ verified
F-22	72.8 s	05	outputs/results.json	runtime_seconds	-	direct	✓ verified
F-23	73,024 orders	05, 08, 09	outputs/results.json	orders_test	-	direct	✓ verified
F-24	log-loss · classwise ECE · daily margin error · COD policy on true p · bootstrap 95% CI	05	src/order_margin/pipeline.py:139-160, 274-275; calibration.py:64-83	-	-	direct	✓ verified
F-25	4 date windows · train · stop · calib · test	05	src/order_margin/pipeline.py:64-71	split_masks keys	-	direct	✓ verified
F-26	expected vs realized daily margin (image)	06	outputs/daily_margin.png via tools/darkmap.py	-	-	direct	manual: Image of the repo's own figure, colour-inverted to dark; legend redrawn because the original legend was painted over.
F-27	122 test days	06	outputs/results.json	test_days	-	direct	✓ verified
F-28	22.5%	06, 08	outputs/metrics.csv	model='class priors', daily_mape_peak (0.2254)	x 100	direct	✓ verified
F-29	4.3%	06, 08	outputs/metrics.csv	model='gbm + isotonic',	x 100	direct	✓ verified

ID	Shown	Slides	Source	Column / row / key	How computed	Type	Check
				daily_mape_peak (0.0431)			
F-30	Class weighting forecasts a loss every day.	06	outputs/ daily_margin.csv	column 'gbm class-weighted, raw' max = -10,259	max < 0	direct	✓ verified
F-31	margin gained against orders lost (image)	06	outputs/ policy_frontier.png via tools/darkmap.py	-	-	direct	manual: Image of the repo's own figure, colour-inverted to dark, then zoomed 1.7x towards the top band; legend redrawn on the right.
F-32	−509,868	06	outputs/policy.csv	policy='block every COD order', margin_change	-	direct	✓ verified
F-33	14,469 orders lost	06	outputs/policy.csv	orders_lost = 14,468.6	rounded	direct	✓ verified
F-34	15.5% of COD hidden	06	outputs/policy.csv	policy='margin rule, gbm + isotonic', blocked_share_of_cod (0.1554)	x 100	direct	✓ verified
F-35	35,389 COD orders	06, 08	outputs/results.json	cod_orders_test	-	direct	✓ verified
F-36	Risk score · same block rate	06	src/order_margin/ pipeline.py:206-218	threshold_for_share(risk _score(pc_c), share)	-	direct	✓ verified
F-37	SALE WEEK · 2025-11-24	07	demo assets/cod_sheet.json; src/order_margin/gener ate.py:30	day, is_peak	-	direct	✓ verified
F-38	609 COD orders (6 rows + 603 more)	07	demo assets/cod_sheet.json; data/orders.csv.gz	cod_orders; COD orders placed on 2025-11-24	-	direct	✓ verified
F-39	102 of 609 COD orders hide COD · 507 keep it	07	demo assets/cod_sheet.json	blocked, kept	102 + 507 = 609	direct	✓ verified
F-40	+602 margin that day	07	demo assets/cod_sheet.json	day_true_gain	sum over hidden orders of conv x E_true[card] - E_true[cod]	direct	✓ verified
F-41	315400, 316055, 316096, 316229, 315744, 316309	07	demo assets/cod_sheet.json, injected into compositions/value.htm l	rows	gain = conv x IF CARD - IF COD; action = HIDE COD if gain > 0	direct	✓ verified
F-42	(consistency check, not on screen)	07	demo assets/cod_sheet.json; outputs/policy.csv	test_window_blocked vs policy 'margin rule, gbm + isotonic' cod_orders_blocked	5,498 = 5,498; 186 rounds = 186	derived	✓ verified
F-43	+ 603 more COD orders	07	F-38	609 - 6 rows	609 - 6 = 603	derived	✓ verified
F-44	63%	08	outputs/policy.csv	margin rule 22,258 / true-probability ceiling 35,479	22,258 / 35,479 = 62.7%	derived	✓ verified
F-45	true p ceiling: +35,479	08	outputs/policy.csv	policy='margin rule, true probabilities (ceiling)', margin_change	-	direct	✓ verified
F-46	+24,371	08	outputs/results.json	margin_minus_risk_rule	-	direct	✓ verified
F-47	95% CI 21,580 to 27,556	08	outputs/results.json	margin_minus_risk_rule _ci95	-	direct	✓ verified
F-48	0.7442	09	outputs/metrics.csv	model='class priors', log_loss	-	direct	✓ verified
F-49	0.6475	09	outputs/metrics.csv	model='logistic, raw', log_loss	-	direct	✓ verified
F-50	0.6469	09	outputs/metrics.csv	model='gbm, raw', log_loss	-	direct	✓ verified
F-51	0.6478	09	outputs/metrics.csv	model='gbm + isotonic', log_loss	-	direct	✓ verified
F-52	0.6181	09	outputs/metrics.csv	model='true probabilities (ceiling)', log_loss	-	direct	✓ verified
F-53	macro-F1 0.290 / 0.294	09	outputs/metrics.csv	macro_f1	-	direct	✓ verified
F-54	ECE 0.0027 → 0.0035	09	outputs/metrics.csv	ece_mean	-	direct	✓ verified
F-55	ECE 0.206 → 0.009	09	outputs/metrics.csv	ece_mean (0.2063, 0.0093)	rounded to 3 decimals	direct	✓ verified
F-56	bars 6 / 767 / 772 / 764 / 1000 px	09	outputs/metrics.csv	(0.7442 - log_loss) / (0.7442 - 0.6181) x 1000	logistic 767, LightGBM 772, LightGBM + isotonic 764	derived	✓ verified
F-57	MIT	10	LICENSE line 1	-	-	direct	✓ verified
F-58	about 2 min	10	README.md line 137	'about 2 minutes on a laptop'	-	direct	✓ verified
F-59	LightGBM · scikit-learn · pandas · Python	10	pyproject.toml dependencies	lightgbm, scikit-learn, pandas	-	direct	✓ verified
F-60	$ make run	10	Makefile	run target	-	direct	✓ verified

2. Automated check run (2026-10-14 14:43)

ID	Expected on slide	Recomputed from repo	Result
F-01	3 chips	-	manual
F-02	true	true	✓ pass
F-03	341,880	341,880	✓ pass
F-04	true	true	✓ pass
F-05	48%	48%	✓ pass
F-06	5	5	✓ pass
F-07	~470	~470	✓ pass
F-08	true	true	✓ pass
F-09	1 in 5	1 in 5	✓ pass
F-10	true	true	✓ pass
F-11	true	true	✓ pass
F-12	true	true	✓ pass
F-13	+22,258	+22,258	✓ pass
F-14	-2,113	-2,113	✓ pass
F-15	true	true	✓ pass
F-16	23	23	✓ pass
F-17	true	true	✓ pass
F-18	186	186	✓ pass
F-19	true	true	✓ pass
F-20	true	true	✓ pass
F-21	true	true	✓ pass
F-22	72.8 s	72.8 s	✓ pass
F-23	73,024	73,024	✓ pass
F-24	true	true	✓ pass
F-25	true	true	✓ pass
F-26	expected vs realized daily margin (image)	-	manual
F-27	122	122	✓ pass
F-28	22.5%	22.5%	✓ pass
F-29	4.3%	4.3%	✓ pass
F-30	true	true	✓ pass
F-31	margin gained against orders lost (image)	-	manual
F-32	-509,868	-509,868	✓ pass
F-33	14,469	14,469	✓ pass
F-34	15.5%	15.5%	✓ pass
F-35	35,389	35,389	✓ pass
F-36	true	true	✓ pass
F-37	true	true	✓ pass
F-38	true	true	✓ pass
F-39	true	true	✓ pass
F-40	+602	+602	✓ pass
F-41	true	true	✓ pass
F-42	true	true	✓ pass
F-43	603	603	✓ pass
F-44	63%	63%	✓ pass
F-45	+35,479	+35,479	✓ pass
F-46	+24,371	+24,371	✓ pass
F-47	21,580 to 27,556	21,580 to 27,556	✓ pass
F-48	0.7442	0.7442	✓ pass
F-49	0.6475	0.6475	✓ pass
F-50	0.6469	0.6469	✓ pass
F-51	0.6478	0.6478	✓ pass
F-52	0.6181	0.6181	✓ pass
F-53	0.290 / 0.294	0.290 / 0.294	✓ pass
F-54	0.0027 → 0.0035	0.0027 → 0.0035	✓ pass
F-55	0.206 → 0.009	0.206 → 0.009	✓ pass
F-56	true	true	✓ pass
F-57	true	true	✓ pass
F-58	true	true	✓ pass
F-59	true	true	✓ pass
F-60	true	true	✓ pass

3. Assumptions register

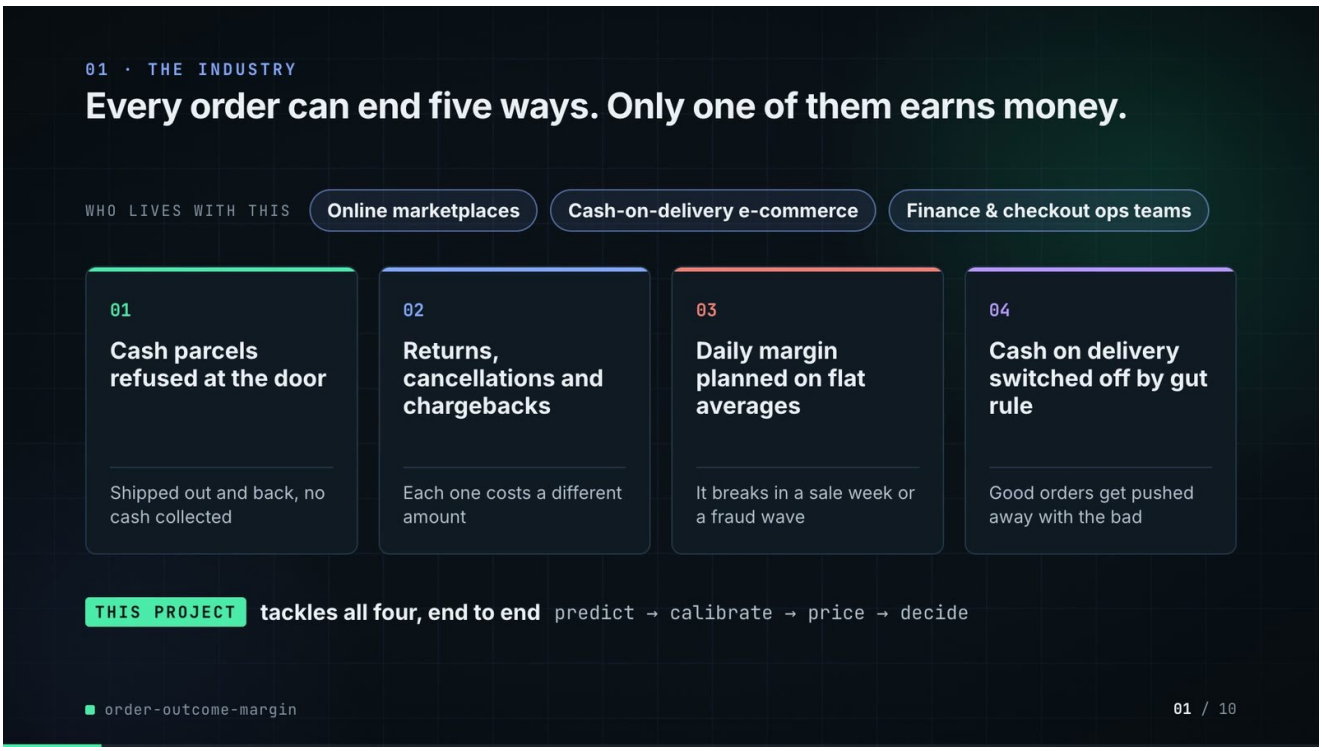
ID	Assumption	Source	Figures affected	Slides affected
A-01	All data is synthetic, produced by a seeded generator sized to the production problem (341,880 orders over 2024-2025, six zones, five categories, COD about 48%). The production data is proprietary.	src/order_margin/config.py:78 (SEED = 20260913); src/order_margin/generate.py; README 'The data'	all figures	slides 01, 02, 03, 04, 05, 06, 07, 08, 09, 10
A-02	Card conversion: 40% of blocked customers younger than 30 days and 60% of the rest switch to a card. It is an input, not measured; with decisions fixed, 20	src/order_margin/config.py:75-76 (COD_TO_CARD_CONVERSION, NEW_CUSTOMER_DAYS); outputs/conversion_sensitivity.	F-13, F-14, F-21, F-32, F-33, F-34, F-39, F-40, F-41, F-44, F-45, F-46, F-47	slides 04, 05, 06, 07, 08

ID	Assumption	Source	Figures affected	Slides affected
	points lower conversion breaks even (+944) and 30 points lower loses 9,714.	csv; README 'Known limitations' (biggest gap)		
A-03	Unit economics are one table per category (gross margin, resale loss, shipping legs, COD/card/wallet fees, a 45 chargeback fee). Real courier rates vary by weight and contract.	src/order_margin/config.py:22-60 (Economics); src/order_margin/economics.py:36-71	F-13, F-14, F-28, F-29, F-30, F-32, F-40, F-41, F-44, F-45, F-46, F-47	slides 04, 06, 07, 08
A-04	Time split: train 2024-01-01..2025-02-28, early stopping 2025-03..04, calibration 2025-05..08, test 2025-09-01..12-31 (73,024 orders, 122 days).	src/order_margin/config.py:67-71; pipeline.py:64-71 (split_masks)	F-20, F-23, F-25, F-27, F-28, F-29, F-35	slides 05, 06, 08, 09
A-05	Outcomes are known 14 days after the order; customer history features only use orders placed at least 14 days earlier, shrunk towards fixed long-run rates.	src/order_margin/features.py:15-19 (RESOLVE_DAYS, PRIOR_STRENGTH, POPULATION_RATE)	F-17	slides 05
A-06	COD policies are scored on the generator's true outcome probabilities under COD and under card (data/truth/order_truth.csv.gz), never on the model that made the decision. The card version of an order is the same order with the payment feature switched.	src/order_margin/pipeline.py:186-201 (counterfactual), :236-245 (score_policy); features.py:with_payment; README 'How it works'	F-13, F-14, F-32, F-40, F-44, F-45, F-46, F-47	slides 04, 06, 07, 08
A-07	LightGBM settings: multiclass, learning rate 0.03, 15 leaves, seed 7, early stopping with 100-round patience on the March-April 2025 window (stopped at 186 rounds); chosen from four settings, never on the test window.	src/order_margin/models.py:71-108; README 'How it works'	F-18, F-19, F-50, F-51	slides 05, 09
A-08	One seed and one test window. The bootstrap interval (1,000 order resamples) covers which orders arrived, not a different year or fraud pattern.	src/order_margin/pipeline.py:268-275; README 'Known limitations'	F-46, F-47	slides 08
A-09	'Sale days' = days flagged is_peak: the November sale week (2025-11-21..29) plus 11.11. The dashed SALE WEEK box on slide 06 marks Nov 21-29 only.	src/order_margin/generate.py:30-31 (SALE_WEEKS, SINGLES_DAY); pipeline.py:74-85 (_daily_errors)	F-28, F-29, F-37	slides 06, 07, 08
A-10	Video rounding: margins to whole units with thousands separators, percentages to 1 decimal (63% and 48% to whole numbers), log-loss to 4 decimals, macro-F1 and ECE as in report.md; '~470 a day' and 'about 2 min' are rounded labels.	this audit (display convention)	F-05, F-07, F-09, F-22, F-28, F-29, F-34, F-44, F-55, F-58	slides 02, 05, 06, 08, 09, 10
A-11	Slide 01 (industry, everyday problems) is general e-commerce knowledge used as context, matched to the README problem statement. It is not a result of this repo and carries no numbers.	author's framing; README 'The problem'	F-01, F-02	slides 01

4. Slide by slide

Slide 01: The industry

0.0-7.2 s · composition `compositions/s01-context.html`



On-screen text (verbatim)

01 · THE INDUSTRY

Every order can end five ways. Only one of them earns money.

WHO LIVES WITH THIS · Online marketplaces · Cash-on-delivery e-commerce · Finance & checkout ops teams

01 Cash parcels refused at the door — Shipped out and back, no cash collected

02 Returns, cancellations and chargebacks — Each one costs a different amount

03 Daily margin planned on flat averages — It breaks in a sale week or a fraud wave

04 Cash on delivery switched off by gut rule — Good orders get pushed away with the bad

THIS PROJECT tackles all four, end to end · predict → calibrate → price → decide

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-01	3 chips	`author's framing; README 'The problem' (finance and operations teams)` · -	-	manual
F-02	4 problem cards	`README 'The problem'` · README lines 12-25	-	✓

Assumptions: A-11

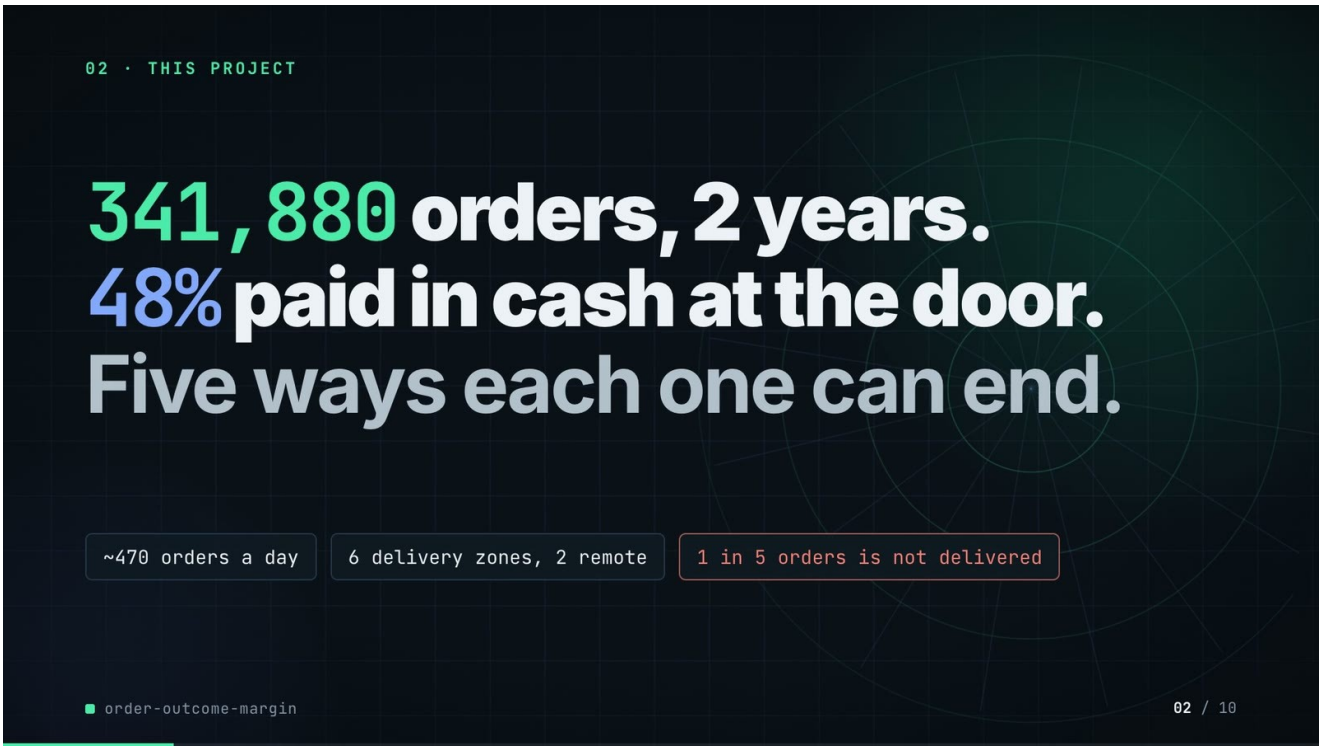
Illustrative, not data:

- The whole slide is context; no repo figures.

To change this slide: Edit the chips (#ctx-c1..3) and cards (#ctx-k1..4) in compositions/s01-context.html.

Slide 02: This project

6.8-11.6 s · composition `compositions/s02-hook.html`



On-screen text (verbatim)

02 · THIS PROJECT

341,880 orders, 2 years. 48% paid in cash at the door. Five ways each one can end.

~470 orders a day · 6 delivery zones, 2 remote · 1 in 5 orders is not delivered

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	341,880 orders	`outputs/results.json` · orders_total	-	✓
F-04	2 years / 2024–2025	`src/order_margin/config.py:70-71` · DATA_START, DATA_END	-	✓
F-05	48%	`data/orders.csv.gz` · payment == 'cod'	share of COD orders = 48.3%	✓
F-06	Five ways each one can end / 5 ways it could end	`src/order_margin/config.py:11` · CLASSES	len = 5	✓
F-07	~470 orders a day	`data/orders.csv.gz` · orders / distinct order days	341,880 / 731 = 467.7, rounded to tens (README: 'around 470 orders a day')	✓
F-08	6 delivery zones, 2 remote	`src/order_margin/config.py:17-18` · ZONES, REMOTE_ZONES	-	✓
F-09	1 in 5 orders is not delivered	`data/orders.csv.gz` · outcome != delivered	1 - 0.798 = 20.2% = 1 in 5 (test window: 20.9%)	✓

Assumptions: A-01, A-10

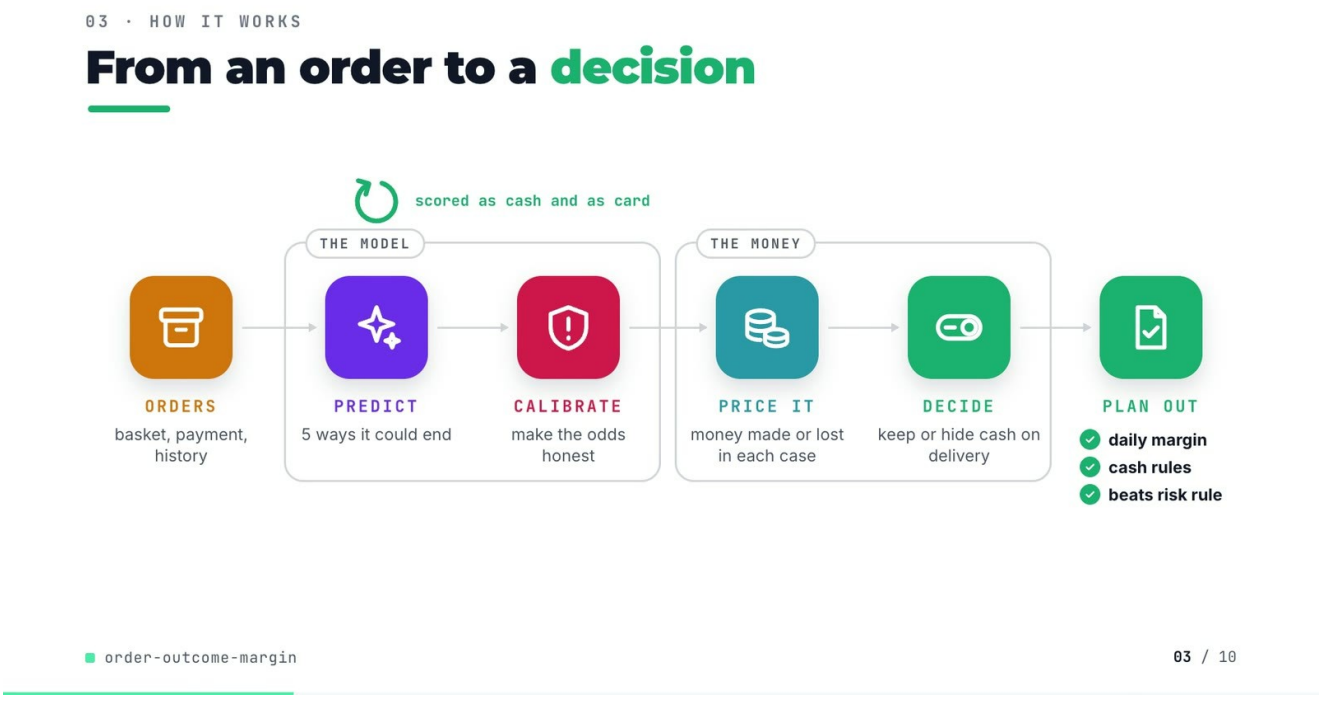
Illustrative, not data:

- Rings in the background are decoration.

To change this slide: Headline spans #hook-l1..l3 and chips #hook-f1..f3 in compositions/s02-hook.html; volumes come from scripts/generate_data.py (make data).

Slide 03: How it works (white chapter, slide A)

11.3-19.3 s · composition `compositions/explainer.html (CONFIG.blocks, CONFIG.groups, CONFIG.loop)`



On-screen text (verbatim)

03 · HOW IT WORKS

From an order to a decision

ORDERS basket, payment, history → [THE MODEL: PREDICT 5 ways it could end → CALIBRATE make the odds honest] → [THE MONEY: PRICE IT money made or lost in each case → DECIDE keep or hide cash on delivery] → PLAN OUT ✓ daily margin ✓ cash rules ✓ beats risk rule

scored as cash and as card (loop over PREDICT)

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-06	Five ways each one can end / 5 ways it could end	`src/order_margin/config.py:11` · CLASSES	len = 5	✓
F-10	scored as cash and as card	`src/order_margin/pipeline.py:193-201` · with_payment(Xt_cod, 'card')	-	✓
F-11	daily margin / cash rules / beats risk rule	`outputs/daily_margin.csv`, `outputs/results.json` · 122 daily rows; margin_minus_risk_rule > 0	-	✓
F-12	CALIBRATE · make the odds honest	`src/order_margin/calibration.py:26-39` · IsotonicCalibrator	-	✓

Assumptions: A-02, A-03, A-06

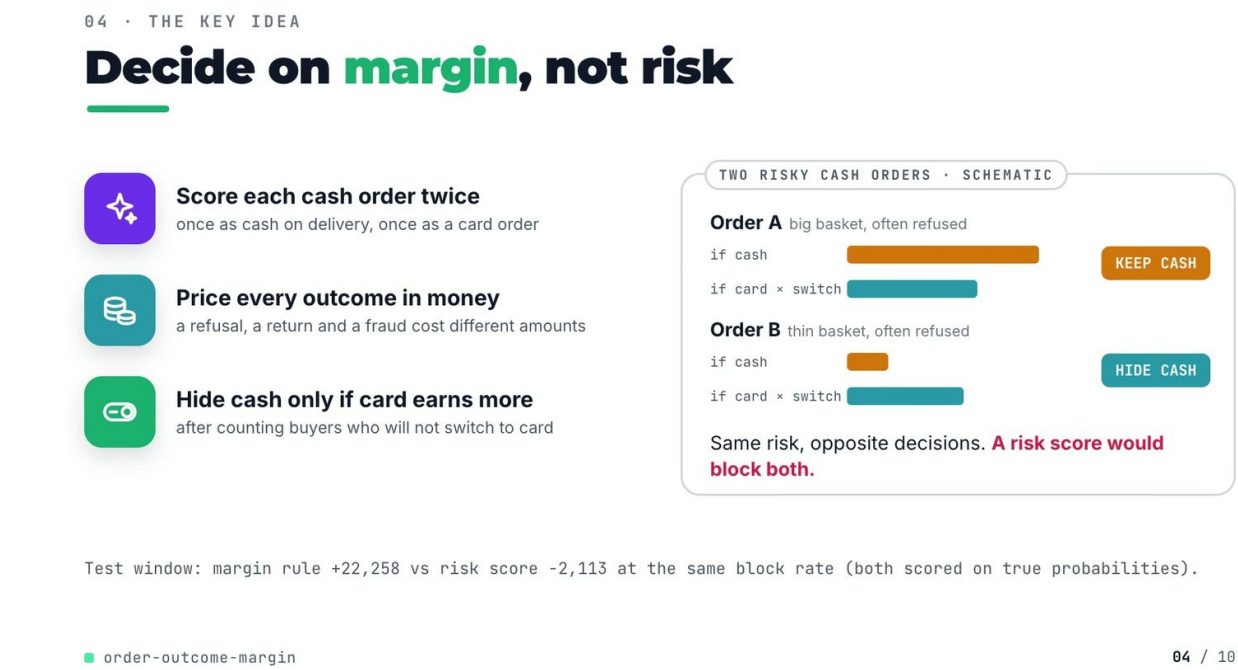
Illustrative, not data:

- Icons and colours are a visual language, not data.

To change this slide: Edit CONFIG in compositions/explainer.html; badge truth comes from F-11.

Slide 04: The key idea (white chapter, slide B)

19.3-26.3 s · composition `compositions/explainer.html (#ex-b ...)`



On-screen text (verbatim)

04 · THE KEY IDEA
Decide on margin, not risk
Score each cash order twice — once as cash on delivery, once as a card order
Price every outcome in money — a refusal, a return and a fraud cost different amounts
Hide cash only if card earns more — after counting buyers who will not switch to card
TWO RISKY CASH ORDERS · SCHEMATIC: Order A big basket, often refused (if cash > if card × switch) KEEP CASH; Order B thin basket, often refused (if cash < if card × switch) HIDE CASH. Same risk, opposite decisions. A risk score would block both.
Test window: margin rule +22,258 vs risk score −2,113 at the same block rate (both scored on true probabilities).

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-13	+22,258	`outputs/policy.csv` · policy='margin rule, gbm + isotonic', margin_change	-	✓
F-14	−2,113	`outputs/policy.csv` · policy='risk score, same block rate as margin rule on calib (14.6%)', margin_change	-	✓

Assumptions: A-02, A-03, A-06

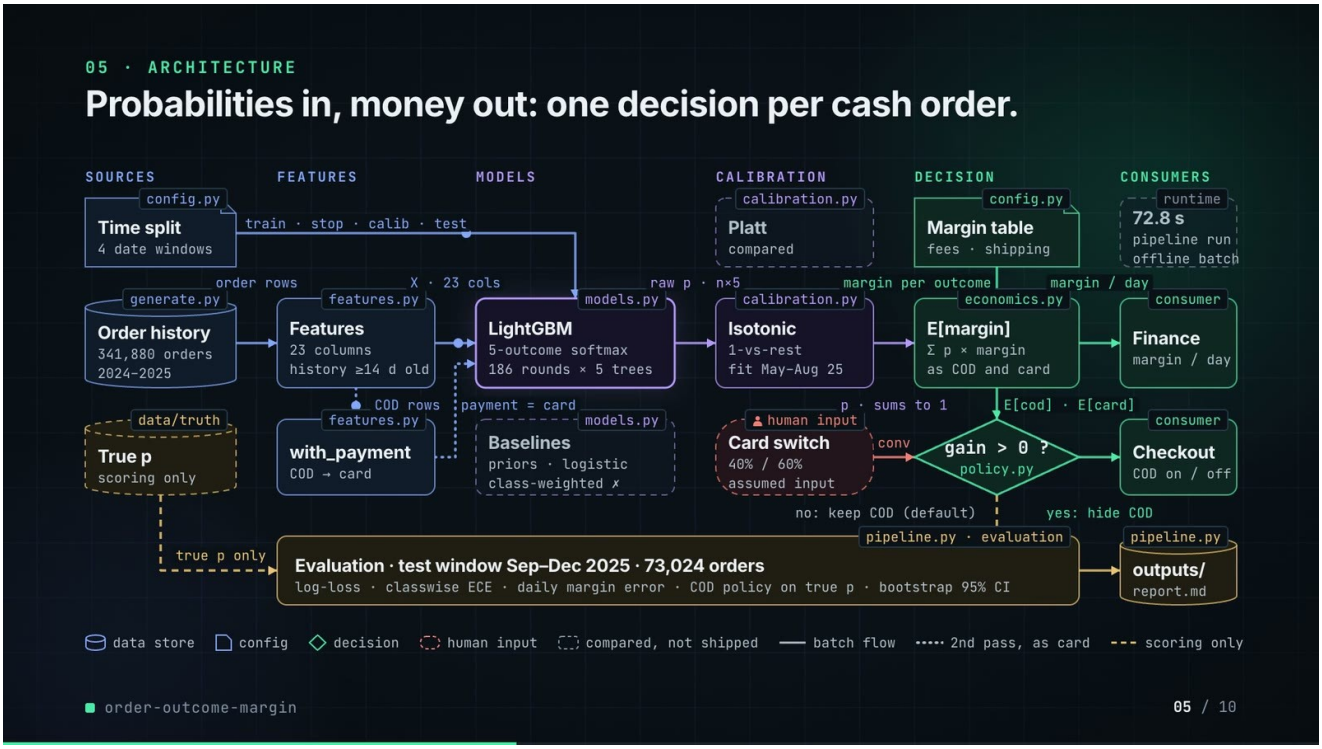
Illustrative, not data:

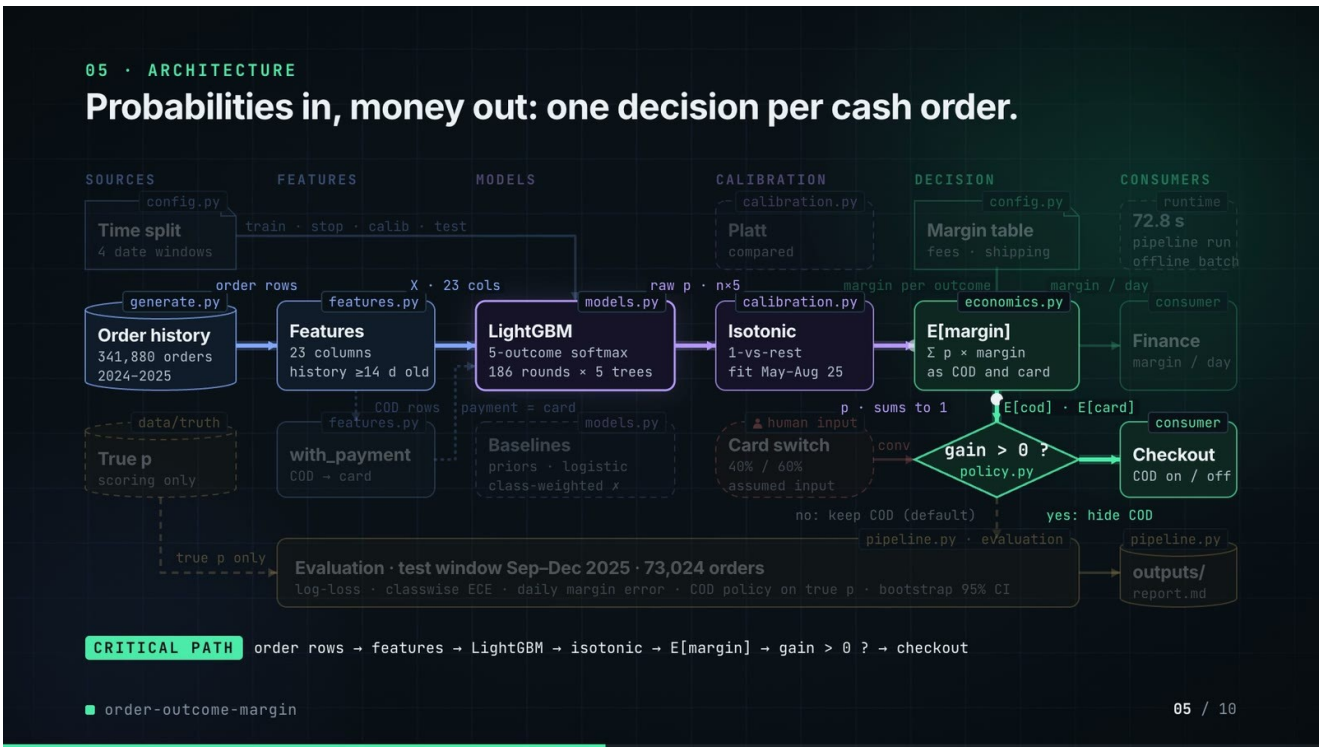
- Orders A and B and their bar lengths are schematic. Real rows with the same pattern are on slide 07 (order 316096: 41% refusal risk, KEEP COD; order 316309: 43%, HIDE COD).

To change this slide: Markup under #ex-b in compositions/explainer.html; the footnote figures are F-13 and F-14.

Slide 05: Architecture

26.0-40.0 s · composition `compositions/s03-architecture.html (N nodes, W wires)`





On-screen text (verbatim)

05 · ARCHITECTURE

Probabilities in, money out: one decision per cash order.

Layers: SOURCES · FEATURES · MODELS · CALIBRATION · DECISION · CONSUMERS

config.py Time split (4 date windows) —train · stop · calib · test→ LightGBM

generate.py Order history (341,880 orders, 2024–2025) —order rows→ features.py Features (23 columns, history ≥14 d old) —X · 23 cols→ models.py LightGBM (5-outcome softmax, 186 rounds × 5 trees) —raw p · n×5→ calibration.py Isotonic (1-vs-rest, fit May–Aug 25) —p · sums to 1→ economics.py E[margin] (Σ p × margin, as COD and card) —margin / day→ Finance

features.py with_payment (COD → card): COD rows in, payment = card back into LightGBM (dotted, 2nd pass)

config.py Margin table (fees · shipping) —margin per outcome→ E[margin]; E[cod] · E[card] → policy.py gain > 0 ? (yes: hide COD → Checkout, COD on / off; no: keep COD (default))

human input: Card switch (40% / 60%, assumed input) —conv→ gain > 0 ?

Ghost boxes (compared, not shipped): Platt (compared), Baselines (priors · logistic, class-weighted ✗); runtime 72.8 s pipeline run, offline batch

data/truth True p (scoring only) —true p only→ pipeline.py evaluation: Evaluation · test window Sep–Dec 2025 · 73,024 orders · log-loss · classwise ECE · daily margin error · COD policy on true p · bootstrap 95% CI → outputs/ (report.md)

Legend: data store · config · decision · human input · compared, not shipped · batch flow · 2nd pass, as card · scoring only

CRITICAL PATH order rows → features → LightGBM → isotonic → E[margin] → gain > 0 ? → checkout

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-03	341,880 orders	outputs/results.json · orders_total	-	✓
F-04	2 years / 2024–2025	src/order_margin/config.py:70-71 · DATA_START, DATA_END	-	✓
F-15	config.py, generate.py, features.py, models.py, calibration.py, economics.py, policy.py, pipeline.py, data/truth	src/order_margin/ · classes and functions	-	✓
F-16	23 columns	src/order_margin/features.py:21-45 · NUMERIC (20) + CATEGORICAL (3)	20 + 3 = 23	✓
F-17	history ≥14 d old	src/order_margin/features.py:15 · RESOLVE_DAYS = 14	-	✓
F-18	186 rounds	outputs/results.json · gbm_rounds	-	✓
F-19	× 5 trees / 5-outcome softmax	src/order_margin/models.py:74-75 · objective multiclass, num_class = N_CLASSES = 5	186 rounds x 5 = 930 trees	✓
F-20	fit May–Aug 25	src/order_margin/config.py:68-69 · EARLY_STOP_END 2025-04-30, CALIB_END 2025-08-31	-	✓
F-21	40% / 60% · assumed input	src/order_margin/config.py:75 · COD_TO_CARD_CONVERSION	-	✓
F-22	72.8 s	outputs/results.json · runtime_seconds	-	✓
F-23	73,024 orders	outputs/results.json · orders_test	-	✓
F-24	log-loss · classwise ECE · daily margin error · COD policy on true p · bootstrap 95% CI	src/order_margin/pipeline.py:139-160, 274-275; calibration.py:64-83 · -	-	✓
F-25	4 date windows · train · stop · calib · test	src/order_margin/pipeline.py:64-71 · split_masks keys	-	✓

Assumptions: A-01, A-02, A-04, A-05, A-06, A-07

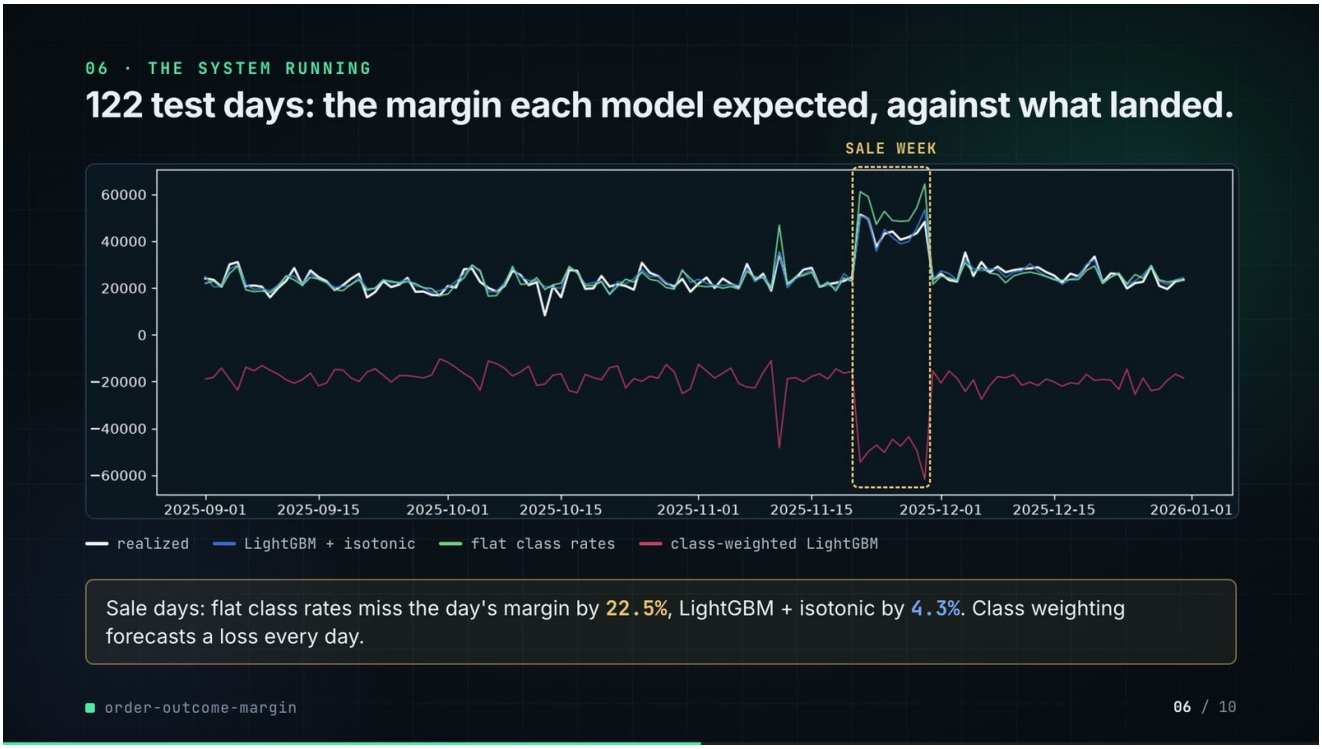
Illustrative, not data:

- Packets moving along the wires show execution order, not timing.
- 'Finance' and 'Checkout' are the consumers the README names (finance close, checkout COD option); the repo writes files, it does not call a checkout service.
- 'offline batch' describes how the repo runs (one CLI run over files); no serving latency is measured, so none is shown.

To change this slide: Node array N and wire array W in compositions/s03-architecture.html.

Slide 06: The system running

39.6-54.2 s · composition `compositions/s04-routes.html; assets/plot-daily.png, assets/plot-frontier.png (from outputs/*.png via tools/darkmap.py)`





On-screen text (verbatim)

06 · THE SYSTEM RUNNING
122 test days: the margin each model expected, against what landed.
SALE WEEK (dashed box) · realized · LightGBM + isotonic · flat class rates · class-weighted LightGBM
Sale days: flat class rates miss the day's margin by 22.5%, LightGBM + isotonic by 4.3%. Class weighting forecasts a loss every day.
Blocking COD: margin gained against orders lost.
rank by margin gain (model) · rank by risk score · rank by margin gain (true p)
Margin rule (15.5% of COD hidden) +22,258 · Risk score (same block rate) −2,113 · Block every COD (14,469 orders lost) −509,868
Decided by LightGBM + isotonic, scored on the generator's true probabilities.

x: orders lost (log) · y: margin change · 35,389 COD orders

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-13	+22,258	outputs/policy.csv · policy='margin rule, gbm + isotonic', margin_change	-	✓
F-14	−2,113	outputs/policy.csv · policy='risk score, same block rate as margin rule on calib (14.6%)', margin_change	-	✓
F-26	expected vs realized daily margin (image)	outputs/daily_margin.png via tools/darkmap.py · -	-	manual
F-27	122 test days	outputs/results.json · test_days	-	✓
F-28	22.5%	outputs/metrics.csv · model='class priors', daily_mape_peak (0.2254)	x 100	✓
F-29	4.3%	outputs/metrics.csv · model='gbm + isotonic', daily_mape_peak (0.0431)	x 100	✓
F-30	Class weighting forecasts a loss every day.	outputs/daily_margin.csv · column 'gbm class-weighted, raw' max = -10,259	max < 0	✓
F-31	margin gained against orders lost (image)	outputs/policy_frontier.png via tools/darkmap.py · -	-	manual
F-32	−509,868	outputs/policy.csv · policy='block every COD order', margin_change	-	✓
F-33	14,469 orders lost	outputs/policy.csv · orders_lost = 14,468.6	rounded	✓
F-34	15.5% of COD hidden	outputs/policy.csv · policy='margin rule, gbm + isotonic', blocked_share_of_cod (0.1554)	x 100	✓
F-35	35,389 COD orders	outputs/results.json · cod_orders_test	-	✓
F-36	Risk score · same block rate	src/order_margin/pipeline.py:206-218 · threshold_for_share(risk_score(pc_c), share)	-	✓

Assumptions: A-02, A-03, A-04, A-06, A-09

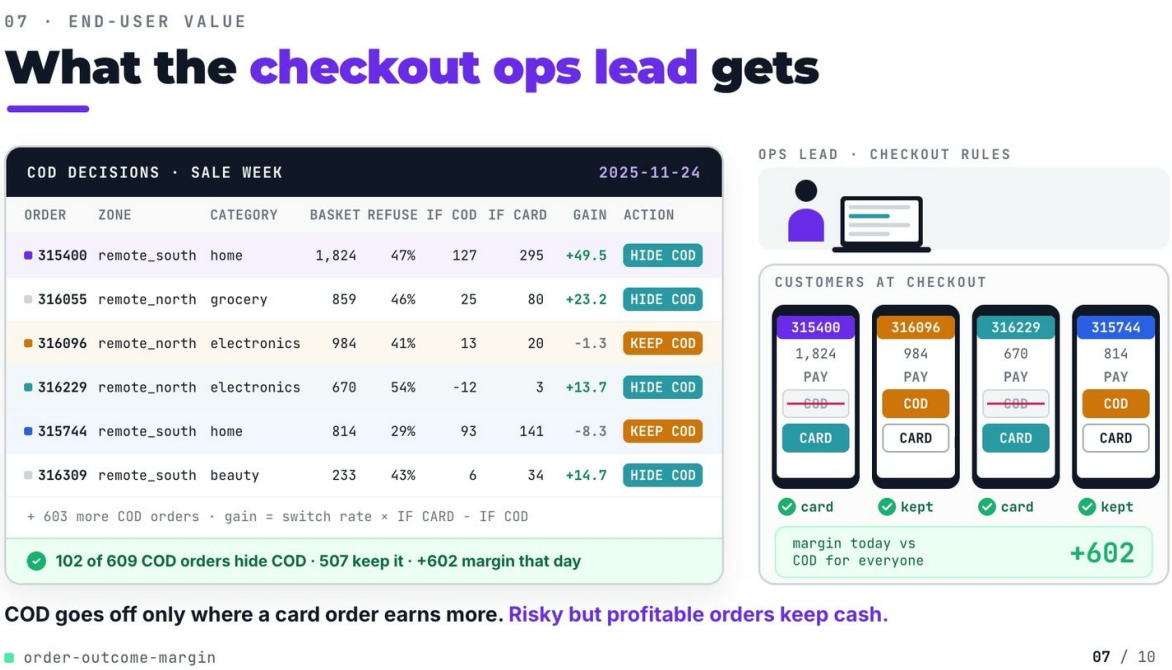
Illustrative, not data:

- Plots are the repo's real figures, colour-inverted to dark; legends are redrawn because the originals were painted over.
- The frontier zooms 1.7x towards its top band; the zoom crops the y tick labels.

To change this slide: Captions and counters in compositions/s04-routes.html; plots come from make run → outputs/daily_margin.png and outputs/policy_frontier.png, then tools/darkmap.py.

Slide 07: End-user value (white)

53.9-66.9 s · composition `compositions/value.html` (SHEET block injected by `tools/inject_sheet.py`)



On-screen text (verbatim)

07 · END-USER VALUE

What the checkout ops lead gets

COD DECISIONS · SALE WEEK · 2025-11-24 — columns ORDER, ZONE, CATEGORY, BASKET, REFUSE, IF COD, IF CARD, GAIN, ACTION

315400 remote_south home 1,824 47% 127 295 +49.5 HIDE COD

316055 remote_north grocery 859 46% 25 80 +23.2 HIDE COD

316096 remote_north electronics 984 41% 13 20 −1.3 KEEP COD

316229 remote_north electronics 670 54% −12 3 +13.7 HIDE COD

315744 remote_south home 814 29% 93 141 −8.3 KEEP COD

316309 remote_south beauty 233 43% 6 34 +14.7 HIDE COD

+ 603 more COD orders · gain = switch rate × IF CARD − IF COD · ✓ 102 of 609 COD orders hide COD · 507 keep it · +602 margin that day

OPS LEAD · CHECKOUT RULES · CUSTOMERS AT CHECKOUT (4 phones: order id, basket, PAY COD / CARD) · card / kept · margin today vs COD for everyone +602

COD goes off only where a card order earns more. Risky but profitable orders keep cash.

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-37	SALE WEEK · 2025-11-24	`demo assets/cod_sheet.json; src/order_margin/generate.py:30` · day, is_peak	-	✓
F-38	609 COD orders (6 rows + 603 more)	`demo assets/cod_sheet.json; data/orders.csv.gz` · cod_orders; COD orders placed on 2025-11-24	-	✓
F-39	102 of 609 COD orders hide COD · 507 keep it	`demo assets/cod_sheet.json` · blocked, kept	102 + 507 = 609	✓
F-40	+602 margin that day	`demo assets/cod_sheet.json` · day_true_gain	sum over hidden orders of conv x E_true[card] - E_true[cod]	✓
F-41	315400, 316055, 316096, 316229, 315744, 316309	`demo assets/cod_sheet.json, injected into compositions/value.html` · rows	gain = conv x IF CARD - IF COD; action = HIDE COD if gain > 0	✓
F-42	(consistency check, not on screen)	`demo assets/cod_sheet.json; outputs/policy.csv` · test_window_blocked vs policy 'margin rule, gbm + isotonic' cod_orders_blocked	5,498 = 5,498; 186 rounds = 186	✓
F-43	+ 603 more COD orders	F-38 · 609 - 6 rows	609 - 6 = 603	✓

Assumptions: A-02, A-03, A-06, A-07

Illustrative, not data:

- Ops-lead figure, laptop, phones and tokens are illustrative; no real people.
- The four phones show four of the six rows (two hidden, two kept), chosen to show both decisions. Order ids are the generator's synthetic ids.
- 'card' under a phone means the order is offered card only; whether that customer converts is the 40/60% assumption (A-02).
- IF COD / IF CARD are the model's expected margins; the day total (+602) is scored on true probabilities, like the repo's policy table.

To change this slide: Re-run tools/export_cod_sheet.py, then tools/inject_sheet.py (--rows / --animate pick the orders); layout in compositions/value.html.

Slide 08: Results

66.6-75.4 s · composition `compositions/s05-results.html (stats array)`



On-screen text (verbatim)

08 · RESULTS · TEST WINDOW SEP–DEC 2025

Margin earned on COD, and a forecast that holds in sale week.

DECISION +22,258 margin, COD margin rule · risk score: −2,113

DECISION 63% of the gain a perfect model gets · true p ceiling: +35,479

FORECAST 4.3% daily margin error, sale days · flat class rates: 22.5%

EVALUATION +24,371 margin rule minus risk rule · 95% CI 21,580 to 27,556

73,024 test orders · 35,389 paid by COD · every policy scored on the generator's true probabilities

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-13	+22,258	`outputs/policy.csv` · policy='margin rule, gbm + isotonic', margin_change	-	✓
F-14	−2,113	`outputs/policy.csv` · policy='risk score, same block rate as margin rule on calib (14.6%)', margin_change	-	✓
F-23	73,024 orders	`outputs/results.json` · orders_test	-	✓
F-28	22.5%	`outputs/metrics.csv` · model='class priors', daily_mape_peak (0.2254)	x 100	✓
F-29	4.3%	`outputs/metrics.csv` · model='gbm + isotonic', daily_mape_peak (0.0431)	x 100	✓
F-35	35,389 COD orders	`outputs/results.json` · cod_orders_test	-	✓
F-44	63%	`outputs/policy.csv` · margin rule 22,258 / true-probability ceiling 35,479	22,258 / 35,479 = 62.7%	✓
F-45	true p ceiling: +35,479	`outputs/policy.csv` · policy='margin rule, true probabilities (ceiling)', margin_change	-	✓
F-46	+24,371	`outputs/results.json` · margin_minus_risk_rule	-	✓
F-47	95% CI 21,580 to 27,556	`outputs/results.json` · margin_minus_risk_rule_ci95	-	✓

Assumptions: A-02, A-03, A-06, A-08, A-09, A-10

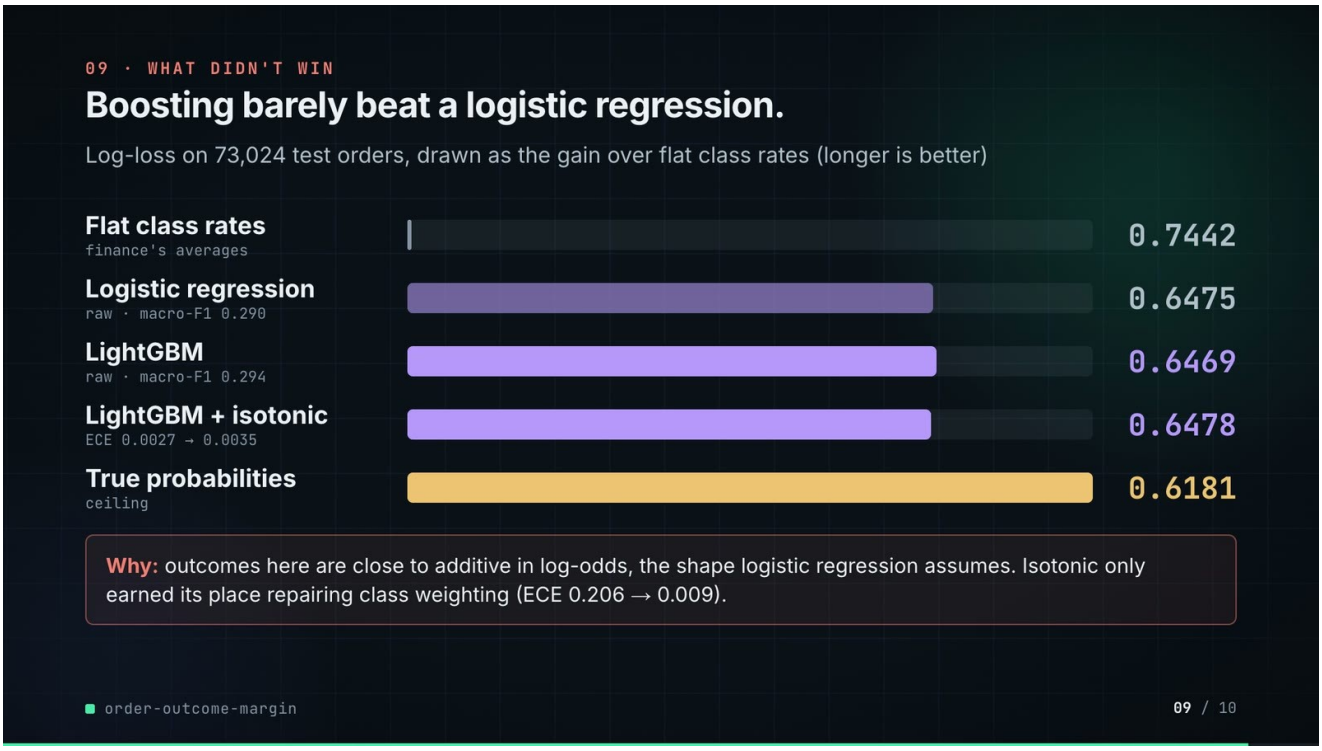
Illustrative, not data:

- Mini bars: cards 1-2 = 22,258 / 35,479; card 3 = 4.3 / 22.5; card 4 is a full bar (no ratio).

To change this slide: The stats array and card text in compositions/s05-results.html.

Slide 09: What didn't win

75.0-81.6 s · composition `compositions/s06-twist.html`



On-screen text (verbatim)

09 · WHAT DIDN'T WIN

Boosting barely beat a logistic regression.

Log-loss on 73,024 test orders, drawn as the gain over flat class rates (longer is better)

Flat class rates (finance's averages) 0.7442 · Logistic regression (raw · macro-F1 0.290) 0.6475 · LightGBM (raw · macro-F1 0.294) 0.6469 · LightGBM + isotonic (ECE 0.0027 → 0.0035) 0.6478 · True probabilities (ceiling) 0.6181

Why: outcomes here are close to additive in log-odds, the shape logistic regression assumes. Isotonic only earned its place repairing class weighting (ECE 0.206 → 0.009).

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-23	73,024 orders	`outputs/results.json` · orders_test	-	✓
F-48	0.7442	`outputs/metrics.csv` · model='class priors', log_loss	-	✓
F-49	0.6475	`outputs/metrics.csv` · model='logistic, raw', log_loss	-	✓
F-50	0.6469	`outputs/metrics.csv` · model='gbm, raw', log_loss	-	✓
F-51	0.6478	`outputs/metrics.csv` · model='gbm + isotonic', log_loss	-	✓
F-52	0.6181	`outputs/metrics.csv` · model='true probabilities (ceiling)', log_loss	-	✓
F-53	macro-F1 0.290 / 0.294	`outputs/metrics.csv` · macro_f1	-	✓
F-54	ECE 0.0027 → 0.0035	`outputs/metrics.csv` · ece_mean	-	✓
F-55	ECE 0.206 → 0.009	`outputs/metrics.csv` · ece_mean (0.2063, 0.0093)	rounded to 3 decimals	✓
F-56	bars 6 / 767 / 772 / 764 / 1000 px	`outputs/metrics.csv` · (0.7442 - log_loss) / (0.7442 - 0.6181) x 1000	logistic 767, LightGBM 772, LightGBM + isotonic 764	✓

F-56 note: The flat-rates bar is a 6 px stub (its gain is zero by definition).

Assumptions: A-01, A-07, A-10

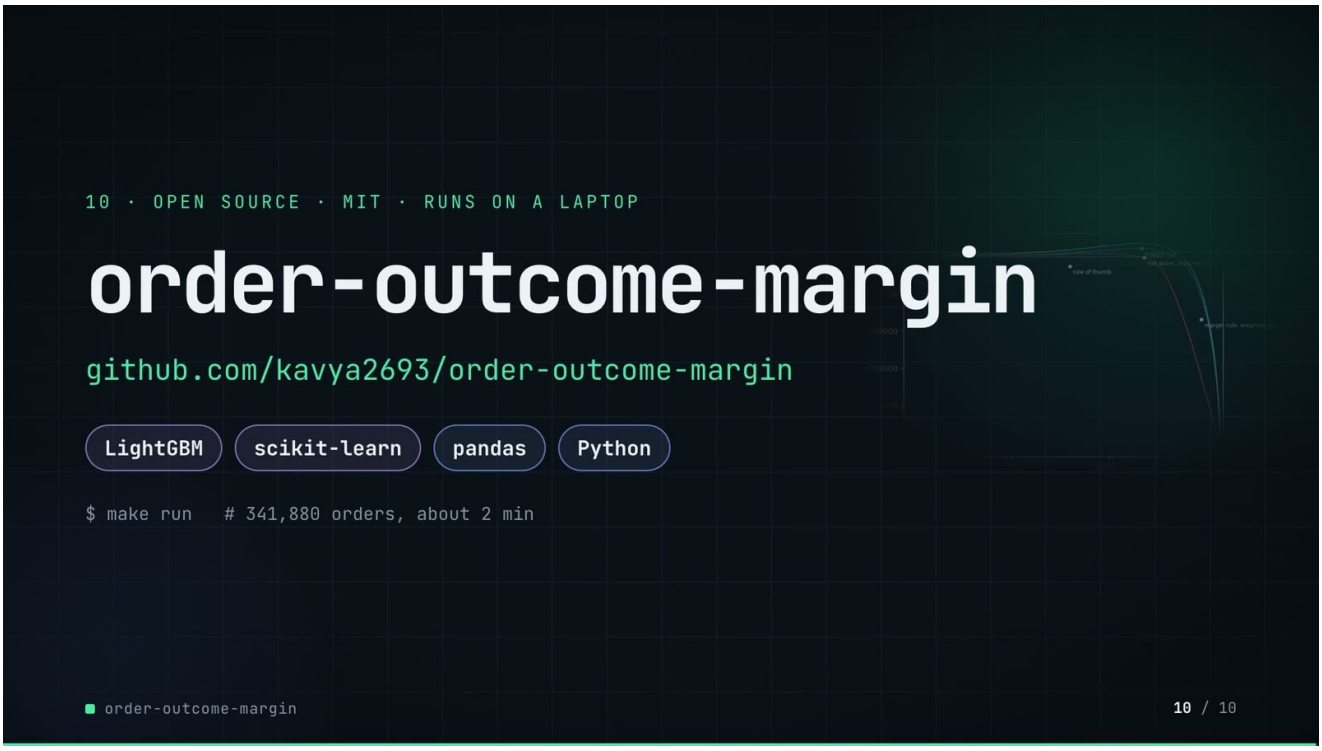
Illustrative, not data:

- Bar lengths: (0.7442 − log-loss) / (0.7442 − 0.6181) × 1,000 px (F-56).

To change this slide: Rows #twist-r1..r5 (bar widths are inline styles) in compositions/s06-twist.html.

Slide 10: End card

81.2-85.6 s · composition `compositions/s07-end.html`



On-screen text (verbatim)

10 · OPEN SOURCE · MIT · RUNS ON A LAPTOP
order-outcome-margin
github.com/kavya2693/order-outcome-margin
LightGBM · scikit-learn · pandas · Python
$ make run # 341,880 orders, about 2 min

Figures on this slide

ID	Shown	Source · key	Calculation	Check
F-57	MIT	`LICENSE` line 1 · -	-	✓
F-58	about 2 min	`README.md` line 137 · 'about 2 minutes on a laptop'	-	✓
F-59	LightGBM · scikit-learn · pandas · Python	`pyproject.toml` dependencies · lightgbm, scikit-learn, pandas	-	✓
F-60	$ make run	`Makefile` · run target	-	✓

Assumptions: A-10

Illustrative, not data:

- Faded plot in the background is the policy frontier figure.

To change this slide: Text in compositions/s07-end.html.

5. Change-impact matrix

Input / assumption	Affected figures	Slides	Files to edit	Re-run
A-01: All data is synthetic, produced by a seeded generator sized to the production problem (341,880 orders over 2024-2025, six zones, five categories, COD about 48%)	all figures	01, 02, 03, 04, 05, 06, 07, 08, 09, 10	`src/order_margin/generate.py`; `src/order_margin/world.py`	make data; make run
A-02: Card conversion: 40% of blocked customers younger than 30 days and 60% of the rest switch to a card	F-13, F-14, F-21, F-32, F-33, F-34, F-39, F-40, F-41, F-44, F-45, F-46, F-47	04, 05, 06, 07, 08	`src/order_margin/config.py:75`	make run; export_cod_sheet.py + inject_sheet.py
A-03: Unit economics are one table per category (gross margin, resale loss, shipping legs, COD/card/wallet fees, a 45 chargeback fee)	F-13, F-14, F-28, F-29, F-30, F-32, F-40, F-41, F-44, F-45, F-46, F-47	04, 06, 07, 08	`src/order_margin/config.py:22-60`	make run; export_cod_sheet.py + inject_sheet.py
A-04: Time split: train 2024-01-01	F-20, F-23, F-25, F-27, F-28, F-29, F-35	05, 06, 08, 09	`src/order_margin/config.py:67-71`	make run
A-05: Outcomes are known 14 days after the order; customer history features only use orders placed at least 14 days earlier, shrunk towards fixed long-run rates	F-17	05	`src/order_margin/features.py:15`	make run
A-06: COD policies are scored on the generator's true outcome probabilities under COD and under card (data/truth/order_truth	F-13, F-14, F-32, F-40, F-44, F-45, F-46, F-47	04, 06, 07, 08	`src/order_margin/pipeline.py`	make run
A-07: LightGBM settings: multiclass, learning rate 0	F-18, F-19, F-50, F-51	05, 09	`src/order_margin/models.py:71-89`	make run
A-08: One seed and one test window	F-46, F-47	08	`src/order_margin/config.py:78`	make data; make run
A-09: 'Sale days' = days flagged is_peak: the November sale week (2025-11-21	F-28, F-29, F-37	06, 07, 08	`src/order_margin/generate.py:30-31`	make data; make run
A-10: Video rounding: margins to whole units with thousands separators, percentages to 1 decimal (63% and 48% to whole numbers), log-loss to 4 decimals, macro-F1 and ECE as in report	F-05, F-07, F-09, F-22, F-28, F-29, F-34, F-44, F-55, F-58	02, 05, 06, 08, 09, 10	the composition file named on each slide	hyperframes render
A-11: Slide 01 (industry, everyday problems) is general e-commerce knowledge used as context, matched to the README problem statement	F-01, F-02	01	`compositions/s01-context.html`	hyperframes render

6. Known differences and caveats

• The COD gain rests on an assumed card conversion (40% / 60%). With decisions fixed, 20 points lower conversion breaks even and 30 points lower loses 9,714 (outputs/conversion_sensitivity.csv, A-02). The README names this the biggest gap.

• Policies are scored on the generator's true probabilities, which only exist because the data is synthetic (A-06). On real data the same comparison needs a randomised checkout holdout.

• Orders lost are counted, not priced: a customer pushed away may not come back (README 'Known limitations').

• 'Sale days' in the 22.5% / 4.3% figures include 11.11 as well as the Nov 21-29 sale week; the dashed box on slide 06 marks the sale week only (A-09).

• Slide 07 rows come from a re-run of the repo's own model for 2025-11-24 (tools/export_cod_sheet.py); the re-run reproduces the repo's 186 rounds and its 5,498 hidden COD orders over the test window exactly (F-42).

• The architecture's 'Finance' and 'Checkout' are the README's consumers; the repo itself writes files (outputs/) and runs as one offline batch. No serving latency exists, so none is shown.

• Slide 01 statements are general industry knowledge (A-11).

Appendix: environment

Item	Value
Commit	`e78d53a391889af780be9f07f440398e04edc426`
Python	3.11.7
lightgbm	4.7.0
sklearn	1.9.1
pandas	3.0.6
numpy	2.4.6

Item	Value
Profile / seeds	data_seed=20260913, lightgbm_seed=7, bootstrap_seed=0
Run time	results.json runtime_seconds = 72.8 s for the pipeline (make run also generates the data first; README says about 2 minutes on a laptop)